

Project Title:	Multi-Scale Overlapping Pattern Tokenization (MSOPT) for Non-Stationary Financial Time Series
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CORE PROBLEM STATEMENT

EXISTING TIME SERIES DEEP LEARNING ARCHITECTURES SUFFER FROM A FUNDAMENTAL PARADIGM FAILURE WHEN APPLIED TO NON-STATIONARY FINANCIAL MARKETS:

1. THE 1D SERIAL POINTWISE BLINDSPOT: Serial models (LSTMs, vanilla Transformers) evaluate scalar price points x_t sequentially. This destroys local visual shape context (inflections, wedges, double bottoms) and causes quadratic computational explosion $O(T^2)$, preventing long historical context processing.

2. THE RIGID UNIFORM PATCHING BOTTLENECK: Modern patch transformers (PatchTST) force rigid uniform sequence slicing ($P=16$). In non-stationary markets, fixed boundaries clip pattern inflections arbitrarily, enforce single-scale rigidity, and lack translation invariance—rendering them fragile during market regime shifts.

3. THE QUANTITATIVE VISUAL GAP: Human quantitative traders analyze market dynamics visually as **multi-scale 2D spatial chart primitives** (micro-spikes, daily consolidations, macro regimes). Existing machine learning paradigms force 1D serial vectors or rigid 1D uniform grids, failing to extract localized multi-scale visual tokens.

1. Executive Summary & Abstract

Financial price series (equities, indices, derivatives) present one of the most challenging domains for deep learning due to extreme non-stationarity, low signal-to-noise ratios, and regime-shifting dynamics. Current models force an unviable tradeoff between point-wise serial processing ($O(T^2)$ computational complexity with zero local shape context) and fixed-patch uniform slicing (boundary clipping distortion with zero multi-scale adaptation).

This research proposes **Multi-Scale Overlapping Pattern Tokenization (MSOPT)**—a novel paradigm that reformulates financial time series modeling. Rather than treating market data as a 1D scalar sequence, MSOPT transforms time series into a **2D Scale-Time Spatial Tensor of multi-scale overlapping receptive fields**. By combining dilated receptive field extraction ($w \in [4, 32], d \in [1, 4]$) with 1D-SAX symbolic discretization (mean + trend slope quantization), MSOPT constructs an interpretable, translation-invariant vocabulary of market pattern tokens ("words").

This proposal outlines the theoretical motivation, literature gap, mathematical framework, validation protocol, and expected departmental contributions of the MSOPT research program at **Universal AI University**.

2. 2025–2026 Literature Landscape & Research Gap Analysis

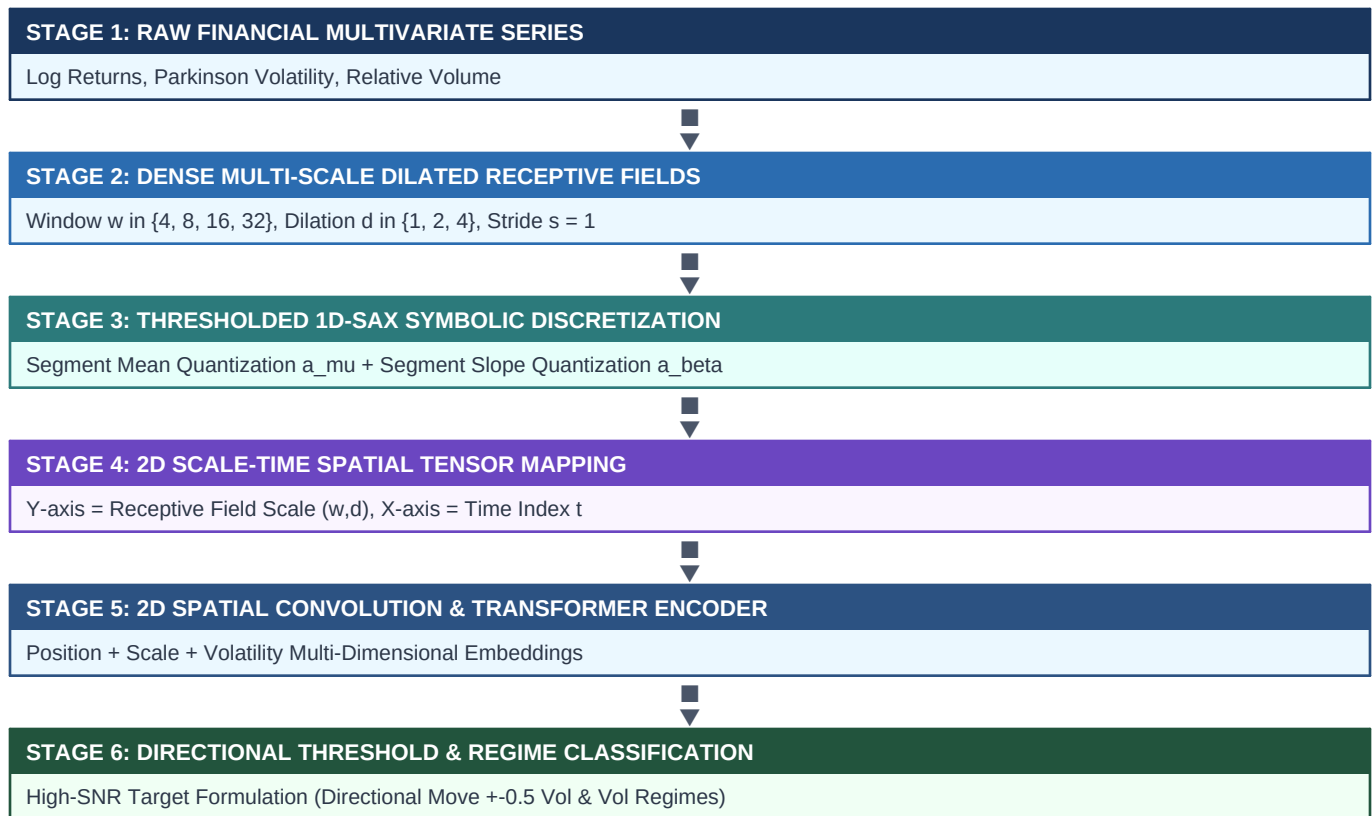
Model Family	Published / Venue	Core Mechanism	Critical Limitation in Financial Series
PatchTST	Nie et al. (ICLR 2023)	Uniform fixed patching ($P=16$, $S=8$) + ViT	Rigid patch length; boundary clipping; zero multi-scale adaptation.
TimesNet	Wu et al. (ICLR 2023)	1D to 2D Tensor Reshaping via FFT periods	Assumes stationary periodicity; fails on non-periodic, regime-shifting financial markets.
BORF	Spinnato et al. (IEEE 2024)	Dilated Receptive Fields + 1D-SAX Discretization	Bag-of-Words loss of temporal sequence order; no end-to-end gradient learning.
TS-BPE	Götz et al. (May 2025/2026)	Byte Pair Encoding subseries merge	Non-overlapping partition ; boundary clipping; not finance-tuned; black-box vocabulary.
DPR	Zhong et al. (May 2026)	Dynamic Pattern Recalibration soft-routing	Recalibration layer on top of tokens; not a token discovery mechanism .
PATK	AAAI-26 (March 2026)	Physics-aware HMM elastic tokenization	Non-overlapping partition ; general-purpose; not human-legible.

The Critical Gap

No existing framework unifies multi-scale dilated receptive field extraction, dense translation-invariant overlapping tokenization ($s=1$), 1D-SAX human-legible codebooks, and 2D scale-time spatial tensor representations within a non-stationary financial modeling paradigm.

3. Proposed Solution: Multi-Scale Overlapping Pattern Tokenization (MSOPT)

MSOPT introduces a 4-stage pipeline designed specifically to solve the non-stationary financial pattern problem:



4. Task Formulation: High-Signal Classification Targets (Fork B Alignment)

To avoid the low signal-to-noise ratio trap of raw point return regression ($\hat{y}_{t+1} \in \mathbb{R}$), where MSE loss forces trivial "flat" predictions), MSOPT evaluates two robust classification tasks:

- **Directional Move Threshold Classification (y_{dir}):**

$$y_{\text{dir}, t} = \begin{cases} +1 & \text{if } R_{t:t+H} > +\delta \cdot \sigma_t \\ -1 & \text{if } R_{t:t+H} < -\delta \cdot \sigma_t \\ 0 & \text{otherwise (Neutral/Noise)} \end{cases}$$

where $\delta = 0.5$ volatility units over horizon $H \in \{1, 5, 20\}$ days.

- **Volatility Regime Shift Classification (y_{vol}):**

$$y_{\text{vol}, t} = \begin{cases} 1 & \text{if } \sigma_{t:t+H} > 1.5 \cdot \bar{\sigma}_{30} \\ 0 & \text{otherwise} \end{cases}$$

Predicting upcoming market volatility expansions and regime breakdowns.

5. Primary Research Questions & Testable Hypotheses

- **RQ1:** Does dense overlapping receptive field tokenization ($s=1$) preserve predictive pattern signal better than single-scale fixed patching (PatchTST) and non-overlapping adaptive partitions (TS-BPE)?
- **H1:** MSOPT will achieve statistically significant out-of-sample Sharpe ratio improvement over PatchTST and TS-BPE due to 100% translation invariance.
- **RQ2:** Does 1D-SAX symbolic discretization improve signal-to-noise ratio in financial returns compared to continuous vector embeddings?
- **H2:** Quantizing local subseries into discrete mean+slope symbols eliminates noise spikes while preserving structural shape primitives.
- **RQ3:** Can a 2D Scale-Time Spatial Grid effectively model inter-scale pattern composition?

- **H3:** 2D spatial convolutions across scale and time dimensions will isolate cross-scale market interactions (e.g., micro-spikes triggering macro-regime breaks) better than 1D temporal convolutions.

6. Rigorous Methodology & Validation Protocol

To guarantee scientific validity and prevent backtest overfitting, MSOPT will be evaluated under a strict protocol:

- **Dataset:** 15+ years of high-liquidity market data (SP500 ETF /SPY, Apple /AAPL, Nasdaq ETF /QQQ, Treasury ETF /TLT) spanning multiple market regimes (2008 Financial Crisis, 2020 COVID shock, 2022 Inflationary bear market, 2023–2026 Tech bull market).
- **Walk-Forward Expanding Window Evaluation:** Initial 5-year training window, expanding annually. Zero lookahead bias.
- **Transaction-Cost Awareness:** All backtests enforce 5 bps slippage and commission costs per trade.
- **Benchmark Comparison:**
 - *Baseline 1:* Standard Technical Features + LightGBM / XGBoost
 - *Baseline 2:* PatchTST (Fixed Patch Transformer)
 - *Baseline 3:* TimesNet (FFT 2D Transformer)
 - *Baseline 4:* TS-BPE (Byte Pair Encoding Transformer)
 - *Baseline 5:* Native BORF + Bag-of-Words Classifier
- **Evaluation Metrics:**
 - Statistical: Directional Accuracy (%), AUC-ROC, Macro F1-Score.
 - Financial: Out-of-Sample Sharpe Ratio, Sortino Ratio, Maximum Drawdown (Calmar Ratio).

7. Project Timeline & Milestones

Phase & Horizon	Milestones & Key Deliverables	Status
PHASE 1 Literature & Blueprint	Deep-dive literature knowledge base across 8 core domains ('research_papers/'), formal Department Research Proposal compilation, and workspace agent memory system setup.	COMPLETED
PHASE 2 Weeks 1 – 2	Formal mathematical specification of 1D-SAX codebook & 2D scale-time spatial grid. Setup of baseline walk-forward backtest protocol environment.	PLANNED
PHASE 3 Weeks 3 – 5	Implementation of MSOPT Tokenizer & 2D Scale-Time Spatial Embedder ('src/tokenizer/'). Controlled ablation studies (MSOPT vs PatchTST vs TS-BPE vs TimesNet vs BORF).	PLANNED
PHASE 4 Weeks 6 – 8	Walk-forward expanding window cross-asset backtesting (SPY, AAPL, QQQ, TLT) with explicit 5 bps transaction costs. Drafting academic manuscript for top-tier submission.	PLANNED

8. Expected Deliverables & Departmental Impact

- **Academic Paper:** A formal research paper targeted for top-tier conference/journal submission (*IEEE PAMI, ICLR, ACM KDD, or Journal of Financial Data Science*).
- **Open-Source Python Framework:** A robust, well-documented time series tokenization toolkit (msopt - t.s) bridging traditional quantitative finance and modern deep learning.
- **Novel Intellectual Property:** A benchmarked, state-of-the-art multi-scale pattern recognition architecture tailored for high-noise non-stationary domains.